

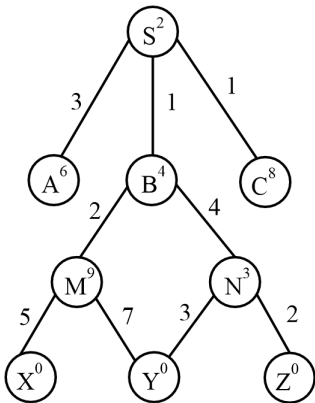
DS&AI

Artificial Intelligence

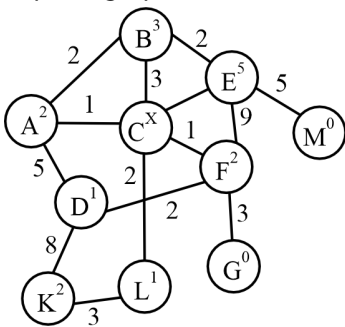
Informed Search

DPP: 1

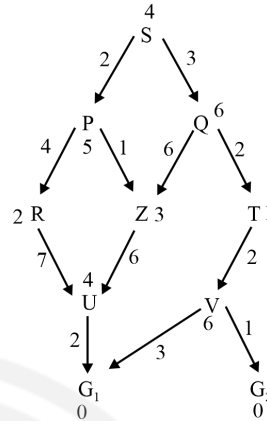
- Q1** Consider the search space depicted in the Figure below. S is the initial state. X, Y, and Z are the states that satisfy the goal test. Use alphabetical order of nodes to break ties. Let P denote the set of states present in the optimal path using A* algorithm and Q represent the set of states present in the optimal path using GBFS. The cardinality of $P \cap Q$ is _____.



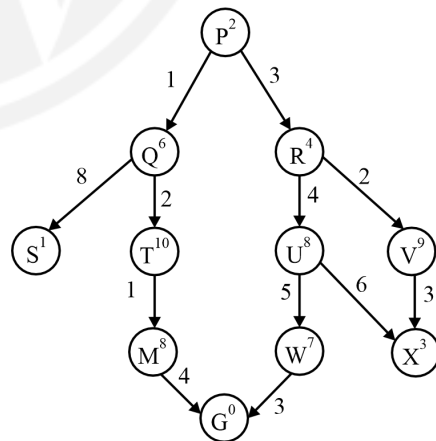
- Q2** What can be the maximum value of 'x' possible to keep the graph consistent?



- Q3** You are performing Weighted A* search on the following graph where the start node is S, and the goal nodes are G1 and G2. The f-value is calculated using the formula: $f(n) = g(n) + w \times h(n)$ where the weight w for the heuristic is set to 3. Calculate the size of the close list after the completion of the entire process.



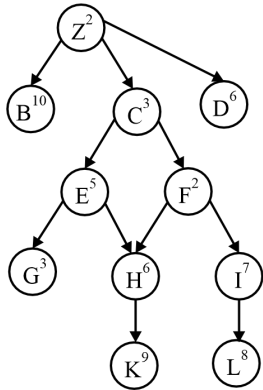
- Q4** The branch factor for a graph is 5 and the depth threshold is 3. The difference between the worst case time and space complexity if Iterative Deepening A* search is used will be _____.
- Q5** What will be the path cost in reaching the goal node G starting from node P, when Breadth First Heuristic Search algorithm is applied on the following graph? Note: Use alphabetical order of nodes to break ties.



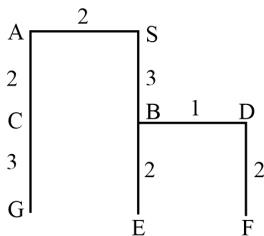
- Q6** In a Hill Climbing problem, we have a graph where the nodes are represented by letters, and each letter is associated with a number based on its position in the alphabet (A = 1, B = 2, C = 3 ..., Z = 26). Z is the start node and G, K, and L are the goal nodes. Determine the code (number)


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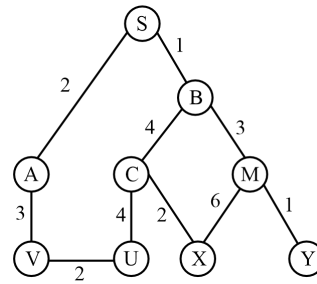
corresponding to the letter on which the Hill Climbing algorithm will stop.



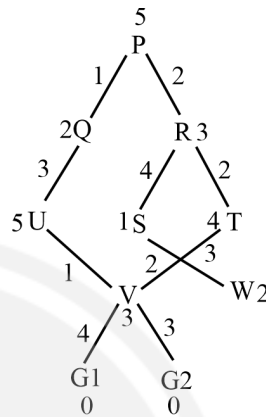
- Q7** Calculate the bound of the following graph where S is the Start node and G and F are goal nodes.



- Q8** Count the number of nodes that will be pruned when branch and bound algorithm is applied on the following graph. S is the start node, U, V, X and Y are end nodes.



- Q9** What will be the threshold value for second iteration when IDA* algorithm is used?



- Q10** Let b denote the branching factor and d denote the maximum depth of a graph. The difference between the worst case time complexity and worst case space complexity on this graph using BFHS is _____.



Answer Key

Q1 **3~3****Q2** **3~3****Q3** **8~8****Q4** **110~110****Q5** **15~15****Q6** **26~26****Q7** **6~6****Q8** **2~2****Q9** **7~7****Q10** **0~0**[Android App](#)[iOS App](#)[PW Website](#)

Hints & Solutions

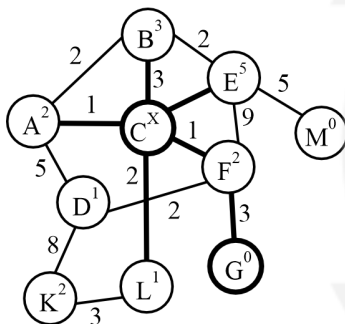
Q1 Text Solution:

	S	B	N	Z
S	0	0	0	
A	9	9	9	
B	5	5	6	
C	9	9	9	
M	∞	12	12	
N	∞	8	8	
X	∞	∞	∞	
Y	∞	∞	8	
Z	∞	∞	7	

$S \rightarrow B \rightarrow N \rightarrow Z$

So, 3

Q2 Text Solution:



Consistent:

$$h(N) \leq C(N, P) + h(P)$$

$C \rightarrow F, L, A, B$

$$h(C) \leq C(C \text{ to neighbour}) + h(\text{height})$$

So,

$$x \leq 6$$

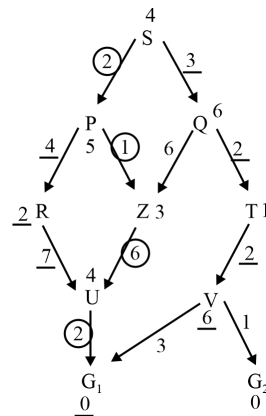
$$x \leq 3$$

$$x \leq 3$$

$$x \leq 3$$

Hence, 3.

Q3 Text Solution:



	S	P	R	Z	Q	T	U	G1
S	0	0	0	0	0	0	0	
P	17	17	17	17	17	17	17	
Q	21	21	21	21	21	21	21	
R	∞	12	12	12	12	12	12	
Z	∞	12	12	12	12	12	12	
T	∞	∞	∞	∞	8	8	8	
U	∞	∞	25	21	21	21	21	
V	∞	∞	∞	∞	∞	25	25	
G1	∞	∞	∞	∞	∞	∞	11	
G2	∞	∞	∞	∞	∞	∞	11	

S P R Z Q T U G1

Close list = 8

Open list = 1

Q4 Text Solution:

$$TC = O(b^d) = 5^3 = 125$$

$$SC = O(bd) = 15$$

$$\text{difference} = 125 - 15 = 110$$

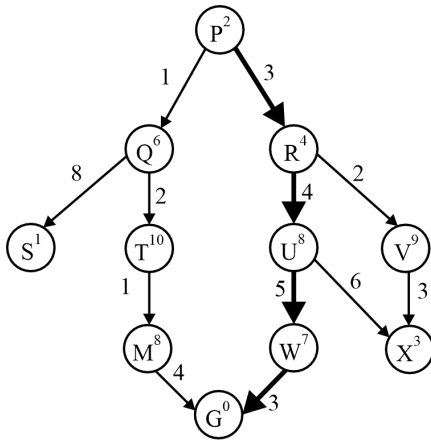
Q5 Text Solution:



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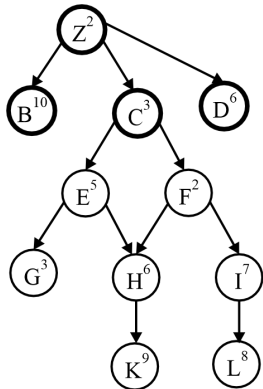


BFHS : P R Q S U X W G

P R U W G

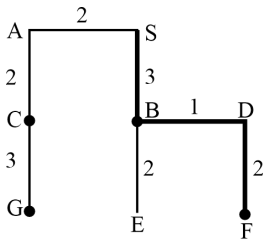
15

Q6 Text Solution:



Z = stop @ z = 26

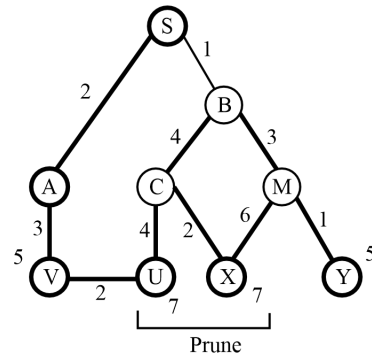
Q7 Text Solution:



S → B → D → F

= 6

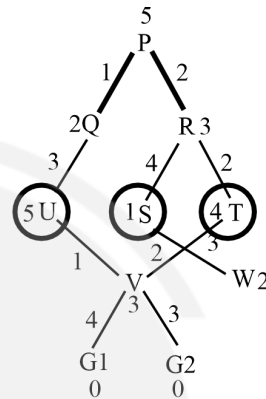
Q8 Text Solution:



Bound = 5

U, X Prune = 2

Q9 Text Solution:



	P	Q	R
P	0	0	0
Q	3	3	3
R	5	5	5
S	∞	∞	7
U	∞	9	
T	∞	∞	8

Next th = min (f value)

7

Q10 Text Solution:

The worst-case time complexity of BFS is $O(b^d)$, where b is the branching factor and d is the maximum depth.

The worst-case space complexity is also $O(b^d)$ due to storing all nodes at the current level.

The difference between the worst-case time and space complexity is 0.

TC = SC = $O(b^d)$



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